HIMANSHU SHARMA

Shimla, Himachal Pradesh | hs2660815@gmail.com | 6230723508 | GitHub | LinkedIn | Portfolio

PROFESSIONAL SUMMARY

Final-year B.Tech. Computer Science student and AI/ML Intern with hands-on experience across the machine learning lifecycle: data cleaning, EDA, feature engineering, model training, evaluation and deployment. Built an AutoML platform (Python, scikit-learn, Streamlit, Docker), a reinforcement learning driving agent in CARLA (SAC), and a fraud-detection model with ROC-AUC 0.96. Skilled in PyTorch, TensorFlow, NLP and LLMs. Seeking a 6-month AI/ML internship.

EDUCATION

Himachal Pradesh Technical University, Hamirpur 2023 – 2027
B.Tech. in Computer Science Engineering
Relevant Coursework: Machine Learning, Deep Learning, Optimization Techniques, Data Structures & Algorithms, DBMS

TECHNICAL SKILLS

Languages: Python, C, C++, SQL
Machine Learning & AI: Machine Learning (classification, regression, clustering), Deep Learning (CNNs, RNNs/LSTMs), Neural Networks, NLP, Transformers, LLMs, Generative AI, Prompt Engineering, Reinforcement Learning (PPO, TD3, SAC)
ML & DL Frameworks: scikit-learn, PyTorch, TensorFlow, Hugging Face
Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn
Deployment & Dev Tools: Streamlit, Docker, Git, GitHub, Jupyter Notebook
Simulation & RL Environments: CARLA Simulator, Gymnasium
Core Concepts: Exploratory Data Analysis (EDA), Data Preprocessing, Feature Engineering, Model Evaluation, Imbalanced Data (SMOTE), Recommendation Systems, AutoML, Model Deployment

EXPERIENCE

AI/ML Intern — Inphora Pvt. Ltd., Mohali Jun 2026 – Aug 2026

- Completed a project-based AI/ML internship covering Python, NumPy/Pandas/EDA, classical machine learning, deep learning with PyTorch (CNNs, RNNs/LSTMs) and Transformer-based NLP.
- Designed and delivered the Problem-Aware AutoML platform (Zoroark-ML) as the internship capstone project.

Data Analytics Job Simulation — Tata Group (Tata iQ) — Forage April 2026

- Conducted GenAI-powered EDA for a financial-services use case, proposed a no-code predictive modeling framework for delinquency risk, and designed an AI-driven collections strategy with ethical and regulatory considerations.

PROJECTS

Problem-Aware AutoML System (Zoroark-ML) | Python, scikit-learn, Streamlit, Docker

- Built an end-to-end AutoML platform with automated dataset profiling, data-quality diagnostics, rule-based preprocessing and algorithm selection based on the inferred problem type.
- Supports classification (Logistic Regression, Random Forest, SVM, KNN) and regression (Linear, Ridge, Lasso, Random Forest Regressor): 8 models across 2 problem types, with results presented in a Streamlit UI.
- Tested the system on 5 datasets and containerized it with Docker for consistent, portable deployment.

Autonomous Driving with Reinforcement Learning | CARLA, Python, Gymnasium, SAC (group of four)

- Compared PPO, TD3 and SAC under identical conditions (200k timesteps each) in CARLA; SAC achieved the highest smoothed episode reward (TD3 stayed slightly negative, PPO was lowest with frequent large drops), so it was trained further to about 1.5M timesteps for the final model.
- Shaped the reward function (speed, progress to destination, lane-center distance, yaw rate, collision and lane-invasion penalties) for a camera-based agent with continuous steer and throttle control; worked in a group of four on environment setup and evaluation.

LLM Game-Theory Agent | Python, LLMs (local), Game Theory (group project, in progress)

- Building, with a group, a four-module local LLM agent for strategic decision-making, with cybersecurity and autonomous-vehicle extensions (ongoing).
- Responsible for the Strategic Reasoning and Solver module.

Credit Card Fraud Detection | Python, scikit-learn, Pandas

- Built a fraud-detection pipeline on 284,807 imbalanced transactions using 20:1 undersampling; compared Logistic Regression, Decision Tree and Random Forest across sampling ratios.
- Selected Random Forest: F1 0.76, precision 0.74, recall 0.78, ROC-AUC 0.96.

Mood-Based Movie Recommender | Python, scikit-learn, Streamlit

- Built a content-based recommender on the TMDB 5000 dataset using CountVectorizer and cosine similarity, deployed as an interactive Streamlit app.

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional Oct 2025
- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate Oct 2025